Protocol fixed (input, splits, operating point, seeds); approved and tagged.

Development on train/validation: whole-card, regularised and tile models; tile model selected; 5 seeds.

Calibration and A4 fusion fitted on validation; parameters committed; signed by all authors.

Test split scored once after a freeze tag; sealed scores hash-checked.

A5 designed from test findings; design recorded before fitting; reported as post-hoc.

Fresh set pre-registered, generated, leakage-checked and evaluated once (confirmatory test of A5).